David Alexandre Fernandes

São Paulo, Brazil
Mobile (WhatsApp) +55 11 9 6778-6118
E-mail: davidalexandrefernandes@outlook.com
LinkedIn: https://www.linkedin.com/in/david-alexandre-fernandes-08b005b4/
Portfolio: https://www.davidalexandrefernandes.com/
Open to relocation across Europe

PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience designing and evolving business-critical software across payments, banking and large-scale e-commerce. Currently developing instant-payment solutions within the Pix ecosystem at Dock using TypeScript, Clean Architecture and AWS. Strong background in Java, Spring Boot, APIs, microservices, distributed systems, enterprise integration, Kubernetes and cloud platforms (AWS, Azure and GCP), with hands-on responsibility for software design, architecture and production reliability.

CAREER OBJECTIVE

To work in Software Engineering, Software Architecture or Artificial Intelligence solutions, contributing to the development of APIs, microservices, cloud platforms, enterprise integrations and applications based on LLMs, RAG and MCP.

PROFESSIONAL EXPERIENCE

Dock - Spread Financial Power | 2026 - Present
Senior Software Engineer

- Develop and evolve new features across Brazil's Pix instant-payment ecosystem, including MED, DICT, SPI, SPD, Scheduled Pix, Automatic Pix, MIP (Matera), QR Code collections, Cash-in and Cash-out.
- Design and implement backend services and payment flows using TypeScript and Clean Architecture, applying Specification-Driven Development (SDD) with OpenSpec and GitHub Spec Kit.
- Design software architecture and technical solutions for new payment features, defining integration flows, component responsibilities and implementation approaches across payment domains.
- Build and evolve cloud-native workloads on AWS using SQS, DLQ, Lambda, Kubernetes and CloudFormation for asynchronous processing and distributed service integrations.
- Apply strong observability and production-diagnostics practices with Datadog, analyzing logs, metrics and application behavior to strengthen reliability and incident investigation.
- Design and evolve data-access flows across relational and non-relational databases supporting transactional, integration and payment-processing workloads.

Banco Bradesco S.A | 2025 - 2026
Senior Software Engineer

- Designed and evolved business-critical APIs and microservices using Java, Spring Boot and Hexagonal Architecture for enterprise banking solutions.
- Provided technical guidance on solution design and engineering decisions, helping define implementation approaches, good practices and technical solutions within the development team.
- Supported application cloud modernization on Microsoft Azure and designed integrations across distributed applications, mainframe systems and critical banking services.
- Implemented and evolved observability and reliability practices using Dynatrace, Open Telemetry, Kibana, Grafana and Prometheus; also developed Angular components integrated with APIs and BFF layers.

Mercado Livre (Mercado Libre) | 2022 - 2024
Software Developer

- Developed and evolved backend systems and scalable APIs using Java and Python for large-scale e-commerce applications.
- Administered Kubernetes environments and applied SRE/platform engineering practices to improve operational stability, troubleshooting and cloud application evolution.
- Provisioned and managed infrastructure as code with Terraform across AWS and GCP, while monitoring logs, metrics and performance with Kibana, Datadog and New Relic.
- Worked with data modelling, DDL creation, database migrations and technical support for system improvements, incident analysis and technology adoption.

Banco Itaú Unibanco | 2020 - 2021
Technology Engineer

- Developed Java features, customizations and plugins for Silk Central and Mobile Center platforms.
- Supported services running on AWS and Docker, including server/application configuration, Oracle database administration and incident resolution.

Cognizant Technology Solutions | 2018 - 2019
Test Automation Engineer

- Developed automated web and mobile tests using Java, JavaScript, Selenium WebDriver, Appium, JUnit and Cucumber.
- Implemented end-to-end automation using Node.js, JavaScript, Postman and Cypress, applying BDD/TDD and supporting defect analysis and test coverage improvements.

Tech Mahindra IT Services | Feb 2018 - Nov 2018
Software Quality Assurance Tester

- Executed end-to-end and integration testing, managed defects in ALM, validated APIs with SOAPUI/JMeter and worked with SQL, Unix and Shell Script for integrated test scenarios.

EDUCATION & TRAINING

Postgraduate Programmer in Software Engineering (MBA)
USP/ESALQ, Brazil | In progress

Postgraduate Degree in Artificial Intelligence Engineering
Faculdade Impacta Digital, Brazil | In progress

Postgraduate Degree in Full Stack Development
FIAP, Brazil | Completed 2026

Postgraduate Programmer in Software Architecture (MBA)
Full Cycle, Brazil | Completed 2025

Bachelor's degree in computer science
Universidade São Judas Tadeu (USJT), Brazil | Completed 2023

Technical Diploma in Information Technology - Systems Development
Centro Paula Souza, Brazil | Completed 2018

TECHNICAL & DIGITAL SKILLS

Backend & Architecture: Java 11+ | Spring Boot | JPA | Maven | Hexagonal Architecture | Clean Architecture | GraphQL API | Quarkus | gRPC API | Microservices | RESTful APIs | Node.js | Express.js | Python | Go | Kafka | Redis | PHP | Laravel | FastAPI | Flask | Flyway | Concurrent Programming | Parallel Programming | Functional Programming
Frontend & Mobile: ReactJS | React Native | JavaScript | TypeScript | HTML5 | CSS3 | Android Studio | Expo Go
Cloud, DevOps & IaC: AWS | GCP | Microsoft Azure | Azure Functions | Kubernetes | Docker | Terraform | CloudFormation | AWS Lambda | SQS | DLQ | Git | GitFlow | Bitbucket | Infrastructure as Code | OpenShift | CI/CD
Databases & Data: SQL | NoSQL | Oracle | MySQL | MongoDB | Databricks | Data Lake
Observability & Monitoring: Datadog | Open Telemetry | Dynatrace | Grafana | Prometheus | Kibana | OpsGenie | New Relic | Power BI | Tableau
Artificial Intelligence: AI Engineering | LLMs | Retrieval-Augmented Generation (RAG) | Model Context Protocol (MCP) | AI Agents | AI APIs | Prompt Engineering | Language Model Integration
Quality & Test Automation: Selenium WebDriver | Appium | JUnit | Cucumber | Mockito | TDD | BDD | Capybara | Postman | SOAPUI | JMeter | End-to-End Testing | Test Automation | ALM | Silk Central | Mobile Center | Device Farm
Systems & Tools: Mainframe | JIRA | DDD | Scrum | Unix | Shell Script | PuTTY | IoT | Kanban | Confluence | Bamboo | Specification-Driven Development (SDD) | OpenSpec | GitHub Spec Kit

PROJECTS

Hortelan - Connected Garden IoT Monitoring Platform
Developed an IoT solution with a web interface for monitoring soil status, battery level, water level, air humidity, temperature, soil moisture and latest irrigation, consolidating data for remote visualization and monitoring.

Web & Mobile Visual Test Automation Solution
Studied, designed and presented a solution for web and mobile visual test automation using JavaScript, Java, Selenium WebDriver, JUnit and Node.js.

Donas do Som - Sony Music Hackathon
Developed a digital platform focused on supporting women's inclusion in the music industry, built with PHP CodeIgniter, HTML5, CSS3, JavaScript, and MySQL.

LANGUAGE SKILLS

Portuguese: Native | English: Intermediate | Spanish: Intermediate